

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack Development Using Mern Stack
Practical – VII

Roll No.: T032	Name: KHUSHI
Class: TYIT	Batch: 2
Date of Assignment: 19-09-26	Date/Time of Submission: 19-09-26

7. MongoDB Basics

a. Install and start MongoDB.

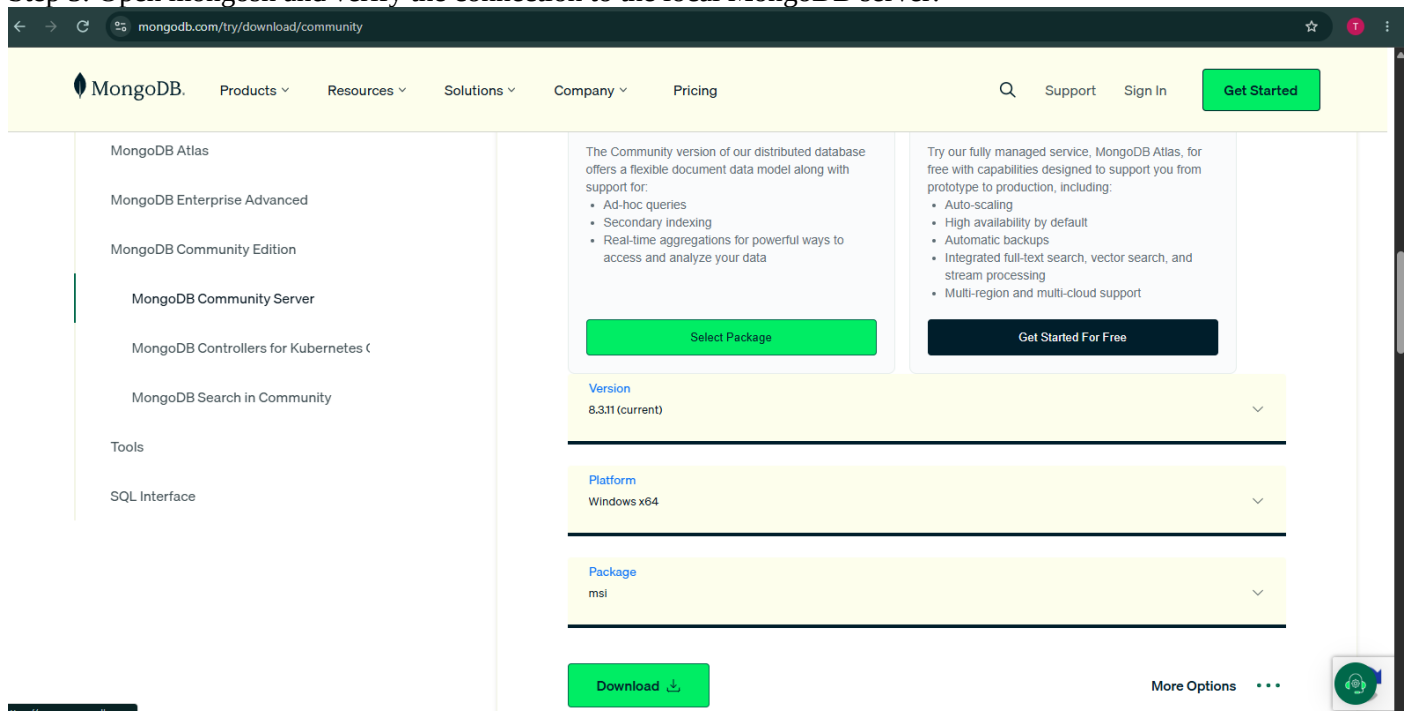
Step 1: Download and install MongoDB Community Server.

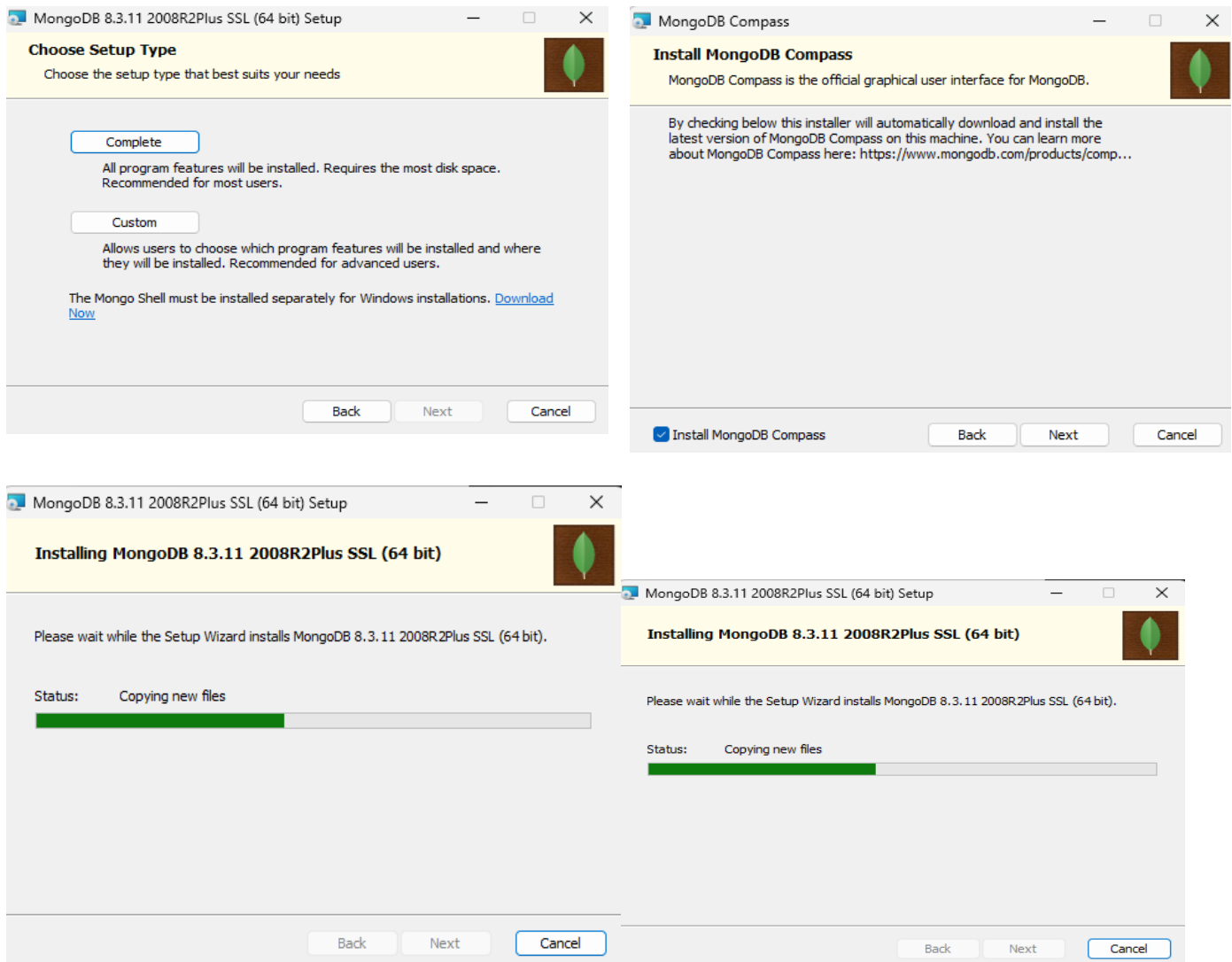
Step 2: During installation, select the option to install MongoDB as a Windows Service.

Step 3: Open Command Prompt or MongoDB Shell (mongosh).

Step 4: Start MongoDB if the service is not already running.

Step 5: Open mongosh and verify the connection to the local MongoDB server.





b. Create a database and collection.

Step 1: Open MongoDB Shell (mongosh).

Step 2: Create or switch to the required database using: `use collegeDB`

Step 3: Create a collection using: `db.createCollection("students")`

Step 4: Verify the collection using: `show collections`

Create Database

Database Name

CollegeDB

Collection Name

Students

☐ Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

➤ **Additional preferences** (e.g. Custom collation, Clustered collections)

Cancel

Create Database

MongoDB Compass - MongoDB practical/CollegeDB.Students

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- MongoDB practical
 - CollegeDB
 - Students**
 - admin

MongoDB practical > CollegeDB > Students

Documents 0 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT DATA EXPORT CODE

Import JSON or CSV file

Insert document

X

Insert Document

To collection CollegeDB.Students

VIEW > {}

```

1 {
2   "rollNo": 101,
3   "name": "Khushi Patel ",
4   "course": "B.Sc IT",
5   "marks": 85
6 }
```

Cancel

Insert

MongoDB Compass - mongodbs/CollegeDB.Students

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongodb
 - CollegeDB
 - Students**
 - admin
 - config
 - local

mongodb > CollegeDB > Students

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1-1 of 1

```

_id: ObjectId('6a9ea14f2bc8b5571c40baa5')
rollNo: 101
name: "Khushi Patel "
course: "B.Sc IT"
marks: 85
```

c. Insert records in MongoDB.

Step 1: Select the database using: use collegeDB

Step 2: Insert one student record using insertOne().

Step 3: Insert multiple student records using insertMany().

Step 4: Verify that the records were inserted successfully.

```
>_MONGOSH
> use("CollegeDB")
< switched to db CollegeDB
> db["Students"].find()
< {
  _id: ObjectId('6aaea14f2bc8b5571c40baa5'),
  rollNo: 101,
  name: 'Khushi Patel ',
  course: 'B.Sc IT',
  marks: 85
}
```

```
> db.students.insertOne({
  rollNo: 102,
  name: "Raj Prajapati",
  course: "B.Sc. IT",
  marks: 92
})
< {
  acknowledged: true,
  insertedId: ObjectId('6aaea1fb71ed35bfb926754a')
}
```

```
>_MONGOSH
> db.students.insertMany([
  {
    rollNo: 102,
    name: "Raj Prajapati",
    course: "B.Sc. IT",
    marks: 92
  },
  {
    rollNo: 103,
    name: "Sachin Mehta",
    course: "B.Sc. CS",
    marks: 78
  },
  {
    rollNo: 104,
    name: "Neha Kulkarni",
    course: "B.Sc. IT",
    marks: 88
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aaea2e671ed35bfb926754b'),
    '1': ObjectId('6aaea2e671ed35bfb926754c'),
    '2': ObjectId('6aaea2e671ed35bfb926754d')
  }
}
```

d. View records from the collection.

Step 1: Select the database using: use collegeDB

Step 2: Display all student records using: db.students.find()

Step 3: Display the records in an easier-to-read format using: db.students.find().pretty()

Step 4: Check the displayed records and verify the inserted data.

```
> db.students.find()
< {
  _id: ObjectId('6aaea1fb71ed35bfb926754a'),
  rollNo: 102,
  name: 'Raj Prajapati',
  course: 'B.Sc. IT',
  marks: 92
}
{
  _id: ObjectId('6aaea2e671ed35bfb926754b'),
  rollNo: 102,
  name: 'Raj Prajapati',
  course: 'B.Sc. IT',
  marks: 92
}
{
  _id: ObjectId('6aaea2e671ed35bfb926754c'),
  rollNo: 103,
  name: 'Sachin Mehta',
  course: 'B.Sc. CS',
  marks: 78
}
{
  _id: ObjectId('6aaea2e671ed35bfb926754d'),
  rollNo: 104,
  name: 'Neha Kulkarni',
  course: 'B.Sc. IT',
  marks: 88
}
CollegeDB>
```

```
> db.students.find({rollNo: 102})
< {
  _id: ObjectId('6aaea1fb71ed35bfb926754a'),
  rollNo: 102,
  name: 'Raj Prajapati',
  course: 'B.Sc. IT',
  marks: 92
}
```

```
> db.students.find({marks: 92})
< {
  _id: ObjectId('6aaea1fb71ed35bfb926754a'),
  rollNo: 102,
  name: 'Raj Prajapati',
  course: 'B.Sc. IT',
  marks: 92
}
```

```
> db.students.find({rollNo: 104})
< {
  _id: ObjectId('6aaea2e671ed35bfb926754d'),
  rollNo: 104,
  name: 'Neha Kulkarni',
  course: 'B.Sc. IT',
  marks: 88
}
> db.students.find({marks: 88})
< {
  _id: ObjectId('6aaea2e671ed35bfb926754d'),
  rollNo: 104,
  name: 'Neha Kulkarni',
  course: 'B.Sc. IT',
  marks: 88
}
```

e. Update and delete records.

Step 1: Select the database using: use collegeDB

Step 2: Update a record using updateOne(), for example: **db.students.updateOne**({rollNo: 101}, {\$set: {marks: 90}})

Step 3: Use **db.students.find()** to verify the updated record.

Step 4: Delete a record using **deleteOne()**, for example: db.students.deleteOne({rollNo: 101})

Step 5: Use db.students.find() again to verify that the record was deleted.

```
> db.students.deleteOne({rollNo: 104})
```

```
< {
  acknowledged: true,
  deletedCount: 1
}
```

```
> db.students.updateOne(
  { rollNo: 101 },
  { $set: { marks: 90 } }
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
> db.students.updateMany(
  { course: "B.Sc.It" },
  { $set: { course: "Information Technology" } }
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
> db.students.updateMany(
  { marks: { $gt: 80 } },
  { $set: { grade: "A" } }
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

```
> db.students.find()
```

```
< {
  _id: ObjectId('6aaaa1fb71ed35bfb926754a'),
  rollNo: 102,
  name: 'Raj Prajapati',
  course: 'B.Sc. IT',
  marks: 92,
  grade: 'A'
}
{
  _id: ObjectId('6aaaa2e671ed35bfb926754b'),
  rollNo: 102,
  name: 'Raj Prajapati',
  course: 'B.Sc. IT',
  marks: 92,
  grade: 'A'
}
{
  _id: ObjectId('6aaaa2e671ed35bfb926754c'),
  rollNo: 103,
  name: 'Sachin Mehta',
  course: 'B.Sc. CS',
  marks: 78
}
CollegeDB>
```

f. Connect MongoDB with Node.js.

Step 1: Create a Node.js project folder and open it in Command Prompt.

Step 2: Initialize the project using: **npm init -y**

Step 3: Install the MongoDB Node.js driver using: **npm install mongodb**

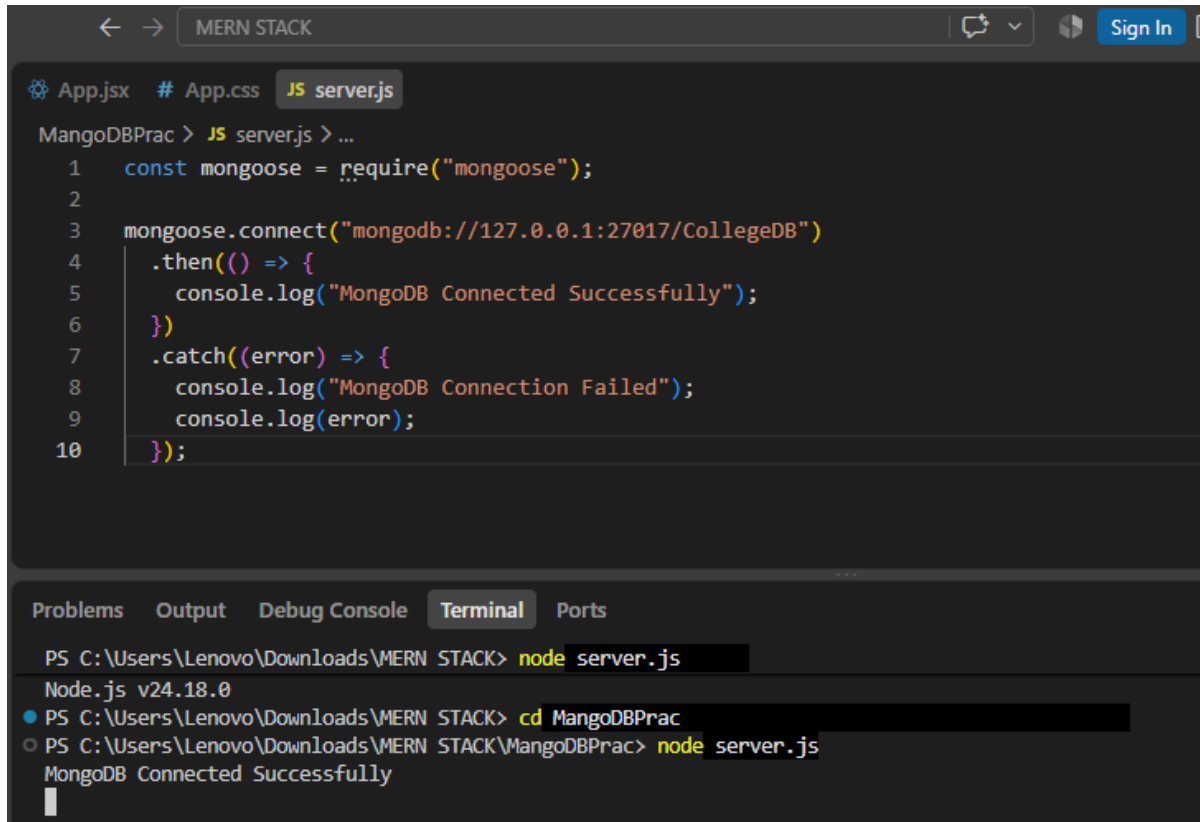
Step 4: Create a JavaScript file such as **server.js**.

Step 5: Import MongoClient and use the local MongoDB connection string.

Step 6: Connect the Node.js application to MongoDB and display a successful connection message.

Step 7: Run the program using: **node server.js**

Step 8: Verify the message "MongoDB Connected Successfully" in the terminal.



The screenshot shows a VS Code editor window titled 'MERN STACK'. The 'server.js' file is open, displaying the following code:

```
MangoDBPrac > JS server.js > ...
1  const mongoose = require("mongoose");
2
3  mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
4    .then(() => {
5      console.log("MongoDB Connected Successfully");
6    })
7    .catch((error) => {
8      console.log("MongoDB Connection Failed");
9      console.log(error);
10   });
```

Below the editor, the 'Terminal' tab is active, showing the command prompt output:

```
PS C:\Users\Lenovo\Downloads\MERN STACK> node server.js
Node.js v24.18.0
PS C:\Users\Lenovo\Downloads\MERN STACK> cd MangoDBPrac
PS C:\Users\Lenovo\Downloads\MERN STACK\MangoDBPrac> node server.js
MongoDB Connected Successfully
```